

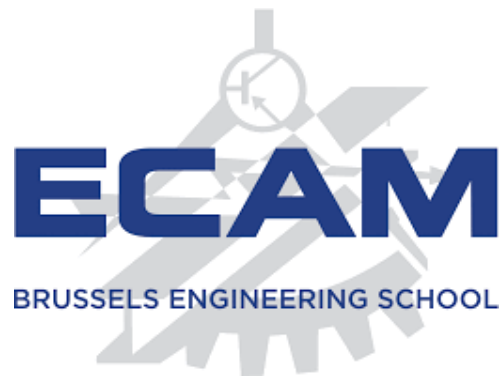


Image Processing 2

Rock-Paper-Scissors

4MEO ECAM

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1 Introduction

For this project, I decided to implement the rock-paper-scissors game using my computer's camera.

2 Objectives

The expected results of this project are the following. First, the camera should start and detect objects of a certain color, blue in this case. Then, it should identify the shape being made. Finally, the user should be able to start a game against the computer or a second player.

3 Challenges

This section presents the main challenges encountered during the project.

3.1 Hand detection

The first challenge, besides starting the camera, was to detect the hand without detecting other objects such as walls, faces, etc. Only the hands should be detected.

3.2 Shape recognition

The second challenge was, once the hand was detected, to recognize the shape it was making.

3.3 Game Manager

The last challenge was to implement a game manager allowing the user to start a game against the computer or a second player.

4 Strategy to overcome theses problems

4.1 Hand detection

The main challenge was not detecting the hand itself because we can easily do that by using some functions from OpenCV library, like `findContours` and `drawContours`. The challenge was to detect the hand only without detecting other objects. There's 2 possible solutions. The first one was to use a blue/green screen to have no interference with the background. The second one is to use bright-colored gloves to isolate the hand from the rest of the frame. I chose the second option because it was easier to implement, as it only required defining an appropriate color mask. This is implemented using the following lines:

```
création du masque de couleur de peau
lower_hand = np.array([180,50,50])
upper_hand = np.array([140,255,255])
mask = cv2.inRange(hsv, lower_hand, upper_hand)
cnts = cv2.findContours(mask, cv2.RETR_TREE, cv2.CHAIN_APPROX_SIMPLE) #détection des contours dans la range du mask
cnts = itertools.chain(*cnts) #permet de récupérer tous les contours de la ligne précédente
```

Figure 1: Color Mask

With this mask, only blue objects are detected.

4.2 Shape Recognition

The second challenge was to recognize the shape being made (rock, paper, or scissors) once the hand had been detected. For this, i decided to use the area of the hand, as the rock has a smaller area than scissors or paper. This was implemented by the following lines :

```
if area < 20000 :
    if color == red :
        player2 = "rock"
    if color == green:
        player = "rock"
elif 20000< area <25000:
    if color == red:
        player2 = "scissors"
    if color == green:
        player = "scissors"
else :
    if color == red :
        player2 = "paper"
    if color == green:
        player = "paper"
```

Figure 2: Area

This method was relatively easy to implement but introduces a subsequent issue: recognition is influenced by the distance between the camera and the hand. As a result, the area thresholds must be adapted depending on this distance.

4.3 Game Manager

The last challenge was implementing a game manager. To play against the computer, the program only needs to recognize one shape, generate a random shape, and compare them. To play with two players, it must recognize two shapes and compare them.

```

if cv2.waitKey(1) == ord('r'): #lancer une partie contre l'ordi
    ordi = random.choice(shapes)
    color = (200,150,100)
    if winner(player, ordi) == player:
        status = "WON"
    elif winner(player, ordi) == ordi:
        status = "LOST"
    else:
        status = "TIE"
if cv2.waitKey(1) == ord('m'): #lancer une partie à 2 joueurs
    if winner(player, player2) == player:
        status = "P1"
    elif winner(player, player2) == player2:
        status = "P2"
    else :
        status = "TIE"

```

Figure 3: Game Manager

With these lines, we can see that pressing "R" starts a game against the computer and when pressing "M" starts a game with 2 players. The function named "winner" determines the winner.

5 Results

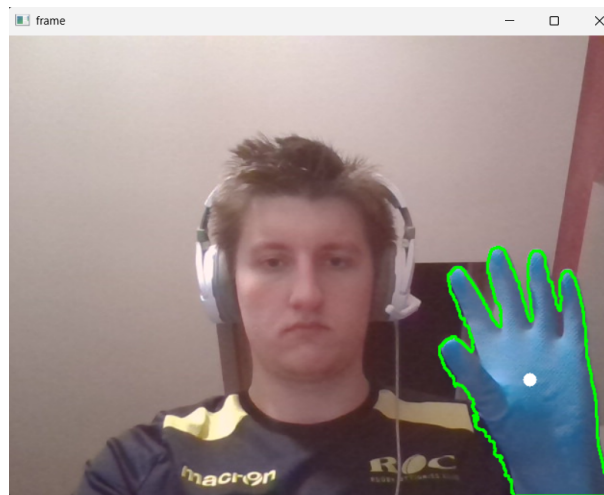


Figure 4: Hand detection

As we can see here, when we start the code, the camera starts and detects only the hand. The game can be started "Q", which will lead to the following menu :

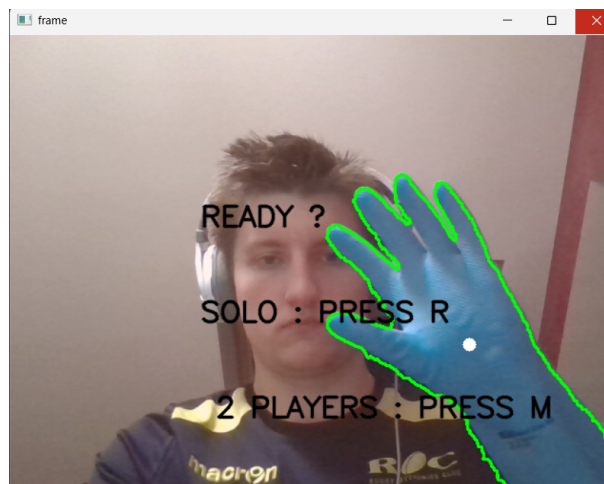


Figure 5: Start Menu

If "R" or "M" is pressed, the winner is displayed on the screen. For example, I pressed "R", which starts a game against the computer :

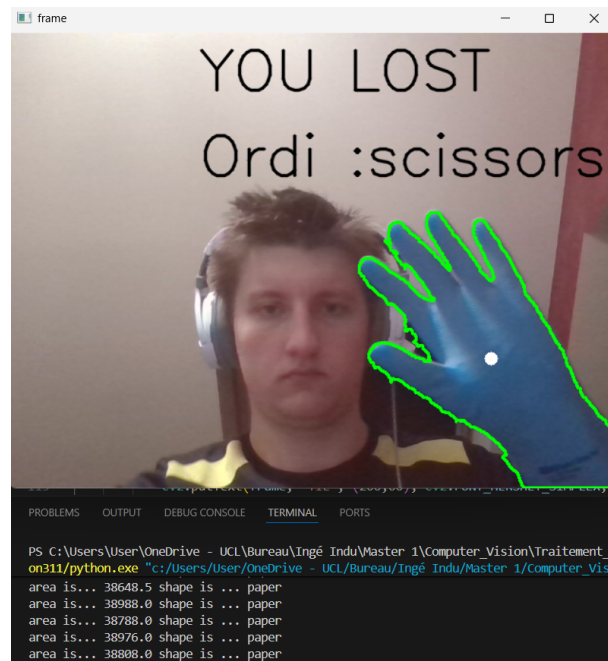


Figure 6: Game

As shown in the terminal I made paper with my hand and the computer ("Ordi" on screen) did scissors. Therefore I lost.

6 Points of improvement

6.1 Hand detection & shape recognition

As mentioned earlier, the user must wear bright-colored gloves for proper hand detection. Something that could be done to improve this would be using a library like MEDIAPIPE, which is trained to recognize and follow hand. This would be much more efficient but you would have to train your code to identify the shape based not on the area of the hand but on the fingers pattern which would be harder to do.

6.2 Game Interface

The game manager could be improved by making the interface more user-friendly. Performance could also be improved (you sometimes have to hold the key for 2-3 seconds for the effect to apply).

7 Video

You can find a video about this project at the following link (<https://youtu.be/XGwhqmudvCc>).

8 AI Usage and Copyright

No generative tools were used to develop this project. ChatGPT 5.2 was used to find spelling or grammar mistakes, as English is not my native language.

The libraries used in this code are the following: Imutils was only used to find the contours of the different objects. OpenCV was used to access the camera, analyze the different shapes detected, and display text on the screen.

I took inspiration from this video (<https://www.youtube.com/watch?v=6AY5p1uC5gM>) from AS-SEMTICA Robotics to indentify the different shapes (lines 41 to 58 in the code).